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Django-based service bidding and procurement information management platform for enterprise services of State Grid companies

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ABSTRACT

With the development of information science, the management methods of various types of information in enterprises are developing towards the direction of intelligence. The improvement of enterprise information management technology in power grid companies is of great significance for improving the management level and efficiency of power grid enterprises. The traditional service bidding and procurement information management for power grid enterprises use paper-based management methods, which often leads to a lack of information. To improve the security and integrity of service bidding procurement information, a service bidding procurement information management platform is proposed for power grid enterprises based on Django. The platform consists of four main components: information comprehensive management module, information loss prevention backup module, access rights management module, and information real-time status management module. It forms a project lifecycle management system for service bidding and procurement information from project requirements, bidding, information management, project progress monitoring, procurement information entry/exit management, and completed project information storage. Under this management platform, the security and standardization of service bidding and procurement information for power grid enterprises can be effectively improved. The work efficiency and management level of relevant departments can also be improved.

CCS CONCEPTS

• **Information systems** → Information systems applications; Enterprise information systems; Enterprise applications; • **Applied computing** → Enterprise computing; Enterprise data management.

KEYWORDS

procurement information, information management, management platform, Django framework

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1 INTRODUCTION

Under the development trend of "Internet +", the information management mode is developing in the direction of intelligence and automation [1]. In the "Top 500 Chinese Enterprises" list in 2022, State Grid Corporation of China ranked first with revenue exceeding 2.9 trillion. As a super enterprise related to China's energy security and the lifeblood of the national economy, State Grid Corporation of China (SGCC) covers 88% of China's land area with its power supply scale.[2] The overall informatization construction of the State Grid of China has been showing a rapid rise, and the investment in informatization by enterprises is also increasing. Various applications such as the power sales system and material management system have gradually become popular among power grid enterprises [3]. Through procurement management, it ensures the normal progress of the production process, and improves the quality of raw materials, controls and reduces all procurement-related costs. The procurement management for SGCC also can provide a strong guarantee for improving product quality and economic benefits. Therefore, strengthening risk management in bidding procurement, identifying risk factors and entities, measuring and evaluating the degree of risk, and determining the best risk control methods are of great significance for ensuring the quality, quantity, and timely supply of electric power materials. Currently, there are certain risks in the bidding and procurement information of

power grid enterprises. The lack of integrity and accuracy of data information will have an impact on the stability of the operation of power enterprises. The main risks include external risks such as unexpected risks, market risks, social risks, and supplier risks, as well as internal risks such as decision-making risks, planning risks, procurement and order risks, transportation risks, acceptance risks, and inventory risks. There are many uncertain risks in the existing information management, such as the lack of tender procurement information and insufficient data security protection. These factors can all become obstacles for power enterprises to systematically complete anticipated projects, impacting the overall arrangement of enterprise development and the expected outcomes of projects.Therefore, it is necessary to develop a relevant management platform for the service bidding and procurement risks of power grid enterprises to achieve scientific management and safe storage of bidding and procurement information.

2 MANAGEMENT REQUIREMENTS ANALYSIS

The service bidding procurement information management platform of a power grid company needs to include processes such as the creation, application, approval, and management of bidding procurement information. In addition, during the design process of the platform, it is necessary to comprehensively consider the storage security of material procurement information, and at the same time achieve access rights settings for confidential information. Through in-depth research on the users of the platform, it is found that corresponding functions need to be designed for the following personnel:

Administrators:

(1) For each registered user of the platform, the accessible range and permissions of users at different levels are set to prevent the disclosure of confidential files.

(2) Propose project requirements for service demand scenarios within the company. Employees conduct further research and submit project applications based on their needs, and the administrator approves the specific details of the project application.

Purchaser/General Staff

(1) Firstly, it is necessary to submit a project application on the platform based on the company's needs and fill in the service bidding and procurement information. The information mainly includes procurement requirements, procurement quantities, project requirements, and other information. (2) After the project is determined, the bidding information is determined, and the file is uploaded to the platform. The platform saves the bidding information according to its characteristics.

(3) After completing the bidding, the document is uploaded to the platform, and can be filled in the details of the winning company, which includes the company name, responsible person information, and the winning amount.

(4) During the process of the project, the account information of the project can be timely processed. It can be filled in the project progress according to the project progress;

In addition, there are warehouse keepers. Their task is to confirm whether the specific content of a service-type bidding procurement project includes goods and register the inbound/outbound goods in detail on the platform.

3 METHODOLOGY

3.1 Information management platform

Based on the needs of service bidding and procurement information management for power grid companies, the structural process of the determined management platform is shown in Figure 1. This platform can realize the entire lifecycle management process of material procurement information from project construction to overall information storage after project completion. In terms of platform functions, the platform mainly includes the following four modules:

(1) Access rights management module

Considering the different confidentiality levels of service bidding and procurement information for power grid services, a management mode for administrator access rights to users was designed during the construction of the platform. When the platform is launched, only an administrator account exists for use by personnel in charge of service bidding and procurement in companies. For any registered user, after registration, it is necessary to apply for the access scope of procurement information based on the individual's work scope. After registering a user's application, the administrator reviews the permissions applied for by the registrant through the access rights management section of the platform, and specifies their information access scope, thereby avoiding the disclosure of confidential information. In addition, during the construction process of this module, the storage of access logs was established to store information access and operation information of different personnel, further ensuring the security of information.

(2) Integrated information management module

This module mainly implements functions including project application, review, project bidding information management, and information saving. For the company's new service bidding procurement needs, it is necessary to apply for project approval on the platform, and the administrator will review the project. If the project passes the review, bidding for the project will be conducted, including generating and saving bid information. If the application fails the review, it will be saved to the failure history database for modification and invocation in the future when there are similar requirements.

(3) Project real-time management module

This module implements the management of approved projects. For projects that have already been approved, the bidding-related documents are saved, and the relevant paper versions are saved to the company's archives. The file information is filled in on the platform. The file needs to be uploaded to the system for intelligent information management. In addition, considering that different bidding procurement projects have certain differences in terms of amount and confidentiality, the platform conducts hierarchical management of projects based on confidentiality registration, winning company, project amount, etc. First, the projects are classified based on its confidentiality, and restricted the scope of access permissions for confidential projects. The classified confidential information is saved to the database according to the project amount range and the winning company. During the information retrieval process, users have set three retrieval methods: project name, project amount range, and project winning company to facilitate users to quickly locate the required information.

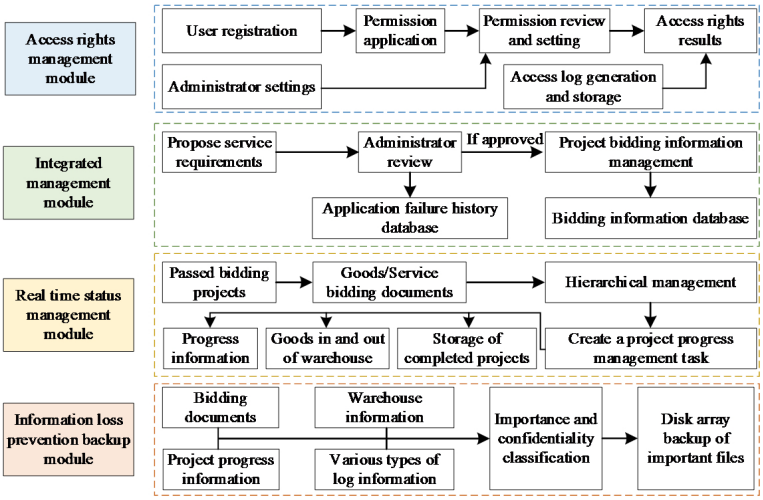

Figure 1: Information management process of power grid enterprise service bidding procurement information management platform

(4) Information loss prevention backup module

The loss of service bidding procurement information may threaten the company's information security and property security. The information that needs to be saved includes electronic bidding documents, warehousing information, various types of log information, and project progress information. To avoid the loss of important information, disk arrays, and multi-disk backups are used for storage. For log files, users can set the save cycle to prevent the log files from occupying a large amount of storage space during the production period.

3.2 Django framework

The management system built in this paper is developed based on the Django framework, which is a Python Web architecture. The 1.0 version of the Django framework was first released in September 2008. The original intention of this framework is to develop a database-driven website that is easy to maintain quickly and easily. Its unique code reuse function supports embedding various components into the entire application framework in a "plug-in" manner, greatly improving the efficiency of application development. The Django framework is mainly used to develop database-driven websites, so it has very powerful database functions, making it easy to associate with different types of databases. The design principle of Django is an MTV (Model-Template-View) pattern that evolved from the MVC (Model-View-Controller) pattern [4, 5]. The pattern comparison between MVC and MTV is shown in Figure 2 [6]. In the MVC design pattern, the Model represents the data storage layer, which defines data tables and processes data. The View represents the graphical display layer, which is the front-end display part of the system. The Controller represents the control layer and is responsible for retrieving data from the Model layer by zeroing the input from the View layer and generating and outputting results based on underlying scripts. In the design mode of MTV, the View layer replaces the Controller layer, and the Template layer replaces the original View layer.

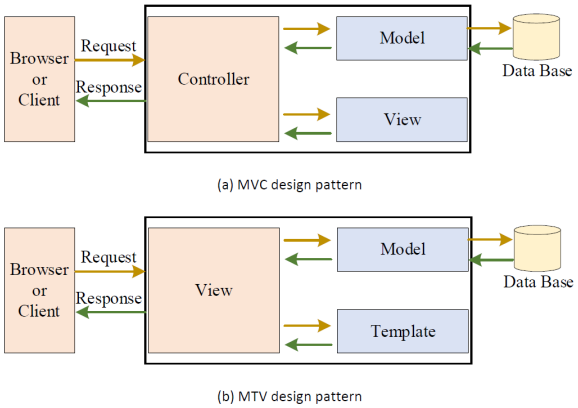

Figure 2: Comparison of design patterns between MVC and MTV

The process of request and response in MTV design mode is as follows:

(1) The user initiates a request to the server through a browser. After receiving the request, the server analyzes it through the business logic layer of View, and sends instructions to the Model layer and Template layer at the same time;

(2) The Model layer interacts with the database and returns the data to the View layer;

(3) After receiving the instruction, the Template layer calls the corresponding template and returns it to the View layer;

(4) After receiving the template and data, the View layer first renders the template (i.e., assigns the corresponding data to the template), then organizes it into a response format and returns it to the browser, which parses it and ultimately presents it to the user.

3.3 MySQL database

The Django framework has an integrated ORM framework that allows users to define model classes directly, without the need for database-oriented programming. This feature makes it easy to support databases such as PostgreSQL, MariaDB, MySQL, Oracle SQLite. All that is needed is to pay attention to the match between the version of Python used by Django and the database version during use. Among various databases, MySQL database has attracted researchers' attention due to its advantages of security, efficiency, and multi-programming language support. During the application process, MySQL database has the advantages of small size, easy maintenance, and low maintenance costs, as well as a variety of database storage engines and powerful built-in functions [7]. It is suitable for the construction of a service bidding and procurement information management platform for power grid companies During the data management process, MySQL database also has features of high response and low latency, allowing for in-depth personalized compilation and management. Based on these advantages of the MySQL database, this platform uses the MySQL database to build a service bidding and procurement information management platform for power grid companies.

In the process of database custom design, reasonable database table settings play an important role in improving data access efficiency. The design process of database tables needs to include user information tables, service bidding and procurement information tables, contract information tables, project progress information tables, goods warehousing information tables, procurement information tables, etc. During the design process of database tables, standardized field information is designed for different data types and security types.

4 REPORT GENERATION SYSTEM

The service bidding and procurement information of power grid companies is important, which needs to be focused on within the company. Related reports of bidding information need to be generated frequently. Considering this demand scenario, a report automatic generation module was added to this platform based on bidding and procurement information management. The process flow is shown in Figure 3. First, the report generation process requires setting report generation options and defining the information contained in the report, including project amount, project progress information, project goods warehousing information, project leader or enterprise information. Secondly, users need to set the update cycle of the report and the time of the first report. The system will automatically calculate the subsequent report generation time based on the first report time. In addition, users can manually and directly generate reports. The process for automatically generating reports in the background of the information management platform is as follows:

(1) In the server, according to the report generation parameter settings set by the user, a .json file is generated and saved for subsequent report generation processes to call.

(2) Information retrieval is performed based on report generation information. During this process, the confidentiality standards for procurement information are strictly followed, and only project information within the set range is retrieved.

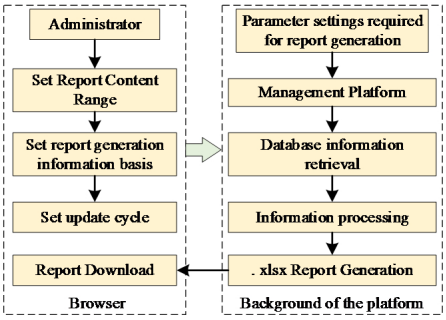

Figure 3: Service bidding and procurement information automatic report generation process

(3) The platform processes the retrieved information, including comprehensive statistics of different types of amounts, as well as the overall progress of different projects. Then, the platform generates detailed information required for different items within the search scope based on the results.

(4) The platform summarizes all information and displays the generated report on the interface.

(5) The user clicks Download and downloads the .xlsx format report file.

5 CONCLUSION

The construction of service bidding and procurement information management platform for power grid companies is of great significance for improving information management efficiency and information security. The research conclusions are as follows:

(1) Based on the needs of power grid companies, this study conducted a demand analysis for platform construction, clarifying the overall architecture of the platform

(2) A power grid company service bidding and procurement information management platform has been developed that integrates an information comprehensive management module, information loss prevention backup module, access rights management module, and information real-time status management module.

(3) The construction of the platform serves as a full lifecycle project management system for bidding procurement projects, including project construction, project progress supervision, project management, and completion of project information preservation.

(4) Based on the reporting requirements of power grid companies, an automatic generation system for bidding and procurement information reports is designed, which can automatically generate service bidding and procurement information reports regularly.

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